

React HandsOn 15:-

Theory:

- Explain about React forms.

React Forms are used to collect user input in a React application. They work similarly to HTML forms but are managed using React's state. Form elements such as text boxes, text areas, checkboxes, radio buttons, and dropdowns are connected to the component's state, allowing React to control their values. This approach makes form handling more efficient and enables features like validation, dynamic updates, and controlled input management.

- Define controlled components.

A Controlled Component is a form element whose value is controlled by the React state. Instead of the browser managing the form data, React stores the input values in the component's state and updates them through event handlers like `onChange`. This provides better control over user input, simplifies validation, and keeps the UI synchronized with the application's data.

Example:

```
<input
  type="text"
  value={this.state.employeeName}
  onChange={this.handleChange}
/>
```

- Explain about various input controls.

React supports all standard HTML form controls:

1. Textbox (`<input type="text">`): Used to accept single-line text input, such as names or email addresses.
2. Textarea (`<textarea>`): Used for entering multi-line text, such as feedback or complaints.
3. Button (`<button>`): Used to submit forms or trigger actions.
4. Checkbox (`<input type="checkbox">`): Allows users to select one or more options.
5. Radio Button (`<input type="radio">`): Lets users select one option from a group.
6. Dropdown (`<select>`): Displays a list of predefined options from which the user can choose.

- Explain about handling forms.

Handling forms in React involves storing form data in the component's state and updating it whenever the user interacts with the form. The `onChange` event is commonly used to capture user input and update the state. This ensures that the displayed values always reflect the current state, making the application responsive and easier to manage.

- Explain about submitting forms.

Form submission in React is typically handled using the `onSubmit` event of the `<form>` element. The event handler usually calls `event.preventDefault()` to prevent the browser's default page reload. The submitted data can then be validated, processed, stored, or sent to a server. In this lab, the `handleSubmit()` method generates a random reference number and displays it in an alert box after the complaint is submitted successfully.

Practical:-

Code:

```
App.css counterapp\... M X JS index.js counterapp\... M JS ComplaintRegister.js U X JS App.js ticketraisingapp\... M # App.css ticketraisingapp\... f
1 import React, { Component } from "react";
2
3 class ComplaintRegister extends Component {
4   constructor(props) {
5     super(props);
6     this.state = {
7       employeeName: "",
8       complaint: ""
9     };
10  }
11  handleChange = (event) => {
12    this.setState({
13      [event.target.name]: event.target.value
14    });
15  };
16  handleSubmit = (event) => {
17    event.preventDefault();
18    const refNumber = Math.floor(Math.random() * 1000000);
19    alert(
20      `Complaint submitted successfully!\nReference Number: ${refNumber}`
21    );
22    this.setState({
23      employeeName: "",
24      complaint: ""
25    });
26  };
}
```

```

    };
    render() {
      return (
        <div className="container">
          <h2>Complaint Register</h2>
          <form onSubmit={this.handleSubmit}>
            <label>Employee Name</label>
            <br />
            <input
              type="text"
              name="employeeName"
              value={this.state.employeeName}
              onChange={this.handleChange}
              required
            />
            <br /><br />
            <label>Complaint</label>
            <br />
            <textarea
              name="complaint"
              rows="5"
              cols="40"
              value={this.state.complaint}
              onChange={this.handleChange}
              required
            />
            <br /><br />
            <button type="submit">
              Submit
            </button>
          </form>
        </div>
      );
    }
  }
  export default ComplaintRegister;

```

```

app.css counterapp\... M  JS index.js counterapp\... M  JS ComplaintRegister.js U  JS App.js ticketraisingapp\... M  X  # App.css tic
1  import ComplaintRegister from "../ComplaintRegister";
2  import "../App.css";
3
4  function App() {
5    return (
6      <div className="App">
7        <ComplaintRegister />
8      </div>
9    );
10 }
11
12 export default App;

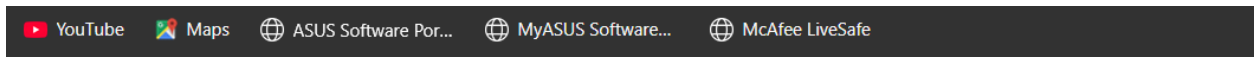
```

```

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...  app.css counterapp\... M  JS index.js counterapp\... M  JS ComplaintRegister.js U  JS App.js ticketraisingapp\... M  # App.css ticketraisingapp\... M  JS
1  import React from "react";
2  import ReactDOM from "react-dom/client";
3  import "../index.css";
4  import App from "../App";
5
6  const root = ReactDOM.createRoot(document.getElementById("root"));
7
8  root.render(
9    <React.StrictMode>
10     <App />
11   </React.StrictMode>
12 );
13

```

Output:



Complaint Register

Employee Name

Complaint

Submit

React App

localhost:3001 says

Complaint submitted successfully!
Reference Number: 873272

OK

Employee Name

Complaint

Submit